

Davide Lamagna

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Software engineer with 5 years of experience, building event-driven pipelines, data-oriented backend systems, and distributed infrastructure. Experienced designing scalable systems and delivering across the full development lifecycle in both industry and academia. Based in Barcelona, known for rapid learning, initiative, and a collaborative mindset.

WORK EXPERIENCE

Barcelona Neural Networking Centre, Computer Architecture

Barcelona; September 2024 – Present

Research Engineer, Agentic AI Systems

- Research group collaboration with UPC, Cisco, and Budapest Technical University under supervision of Prof. Albert Cabellos.
- Leading new research on **agentic AI** and **emerging protocols**, investigating implications on **future internet infrastructure**.
- Lead author on **ACM SIGCOMM** workshop: “Towards Infrastructure-Aware **Traffic Modelling of Multi-Agent Systems**”
- Engineered a **distributed multi-agent system** using Python and Docker, processing and profiling network-level traffic from agentic workloads, with results served to downstream monitoring applications via Prometheus and Grafana.

Research Engineer, Kubernetes

- Technical Support for **IEEE** published paper: "On the Stability of the Kubernetes Horizontal Autoscaler Control Loop".
- Contributing to follow-up paper in June 2026: “**Context-Aware Scaling in Kubernetes**” resource allocation and HPA.
- Prototyped a **decentralised Kubernetes system** using **internodal gossip**, functioning independently of the control plane.

S&P Global, Financial Risk Analytics

London, Barcelona; June 2021 – Present

Software Engineer, Technical Product (Oct 24 – Present; Part-time, Barcelona)

- Developed an end-to-end **regression testing pipeline**, automated via **GitLab CI/CD**, greatly reducing manual QA overhead.
- Contributed to a data store migration to a relational database architecture, working hands-on with **PostgreSQL** and Amazon **Athena** for structured data storage and querying, improving data reliability and query performance across the platform.
- Architected an event-driven **monitoring platform** on **AWS**, integrating real-time metrics through Grafana dashboards.
- Contributed to the design of climate risk frameworks, including top-down and bottom-up macroeconomic scenario features.

Software Engineer, Customer Success (Aug 22 – Aug 24; Full-time, London)

- Built Python tools to close product gaps and streamline internal operations; selected features were integrated into core offering.
- Led agile cycles to develop high-demand features during a **rapid build** phase; oversaw **cross-functional delivery** to clients.
- Orchestrated end-to-end onboarding of **4 buyside and sellside clients** over the year, collaborating closely with relevant SMEs.
- Conducted applied research projects to **automate operational models** using both in-house and third-party technologies
- Led the business’ first-ever **SOC2 audit**, establishing a secure, robust, and fully compliant foundation for long-term scale.

Software Engineer, Summer intern, IHS Markit (Jun – Sep 2021; Full-time, London)

- Supported global Risk as a Service operations; contributed Python and Unix code to embed **DevOps** in daily operations.

Tuition

London; April 2018 – January 2023

Part time, self-employed

- Mathematics tuition for students studying 11+, GCSE, and A Level Exams. University/early careers **mentoring and coaching**.

EDUCATION

Universitat Politècnica de Catalunya, FIB

Barcelona; September 2024 – June 2026

- MEng Innovation and Research in Informatics; specialising in Computer Networks and Distributed Systems. (GPA 8.1/10)
- Awarded full research scholarship with the Computer Architecture department, focused on Kubernetes and Agentic Systems.
- Decentralised Systems (Go/Python), Cloud Computing (AWS), Concurrency/ Functional Programming (Java/C/Erlang/Cuda).
- Information Retrieval (Python/Elasticsearch), covering indexing, search ranking, and recommendation algorithms.

Bocconi University

Milan; September 2019 – June 2022

- BSc Economics, Management, and Computer Science. (Final grade 105/110)
- Modules including Stochastic Modelling, Econometrics, Advanced Programming, Machine Learning, Blockchain Technology.

Blockchain Technology: The Next Stage of The Music Industry’s Evolutionary Path? (Grade distinction)

- Bachelor dissertation on the structural failures of royalty distribution across CMOs and streaming platforms, examining how fragmented metadata and multi-intermediary payment chains result in delayed and inaccurate payments to rightsholders.
- Evaluated blockchain-based alternatives for copyright data infrastructure, including decentralised ledger design, smart contract royalty automation, assessing feasibility against scale, adoption, and industry incumbents.

Nova School of Business and Economics (Erasmus)

Lisbon, January – July 2022

- Electives in Modelling and Optimisation, Advanced Statistics, and Contemporary History.

LANGUAGES AND DEVELOPMENT EXPERIENCE:

- **Languages:** English, Italian (*bilingual*); Spanish (*C1*); French (*B2*); Portuguese, Catalan (*Basic*)
- **Programming: Advanced:** Python (async, multiprocessing), Bash; **Intermediate** PostgreSQL, R, Java, C, Javascript, Go, CUDA
- **DevOps and Cloud:** experience with Kubernetes, AWS, Docker, Git, CI/CD, REST API, Service mesh, Prometheus, Grafana.

ADDITIONAL INFORMATION:

- Freelance developer and research consultant for early-stage startups; **AI prototyping**, and **cloud native architecture** design.
- Historical triathlete and cross-country runner, current passionate weekend rock climber. Recreational guitar player.